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THE CHIP LANDSCAPE

GEOGRAPHICAL DISTRIBUTION OF WAFER FABRICATION CAPACITY

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The chip landscape: Geographical distribution of wafer fabrication capacity



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Abstract

Enhancing the resilience of the complex semiconductor value chain requires better data. This paper strengthens the evidence base for semiconductor policy by mapping global semiconductor wafer fabrication capacity by economy, process node density, process technology and business model, building on the newly assembled OECD Semiconductor Production Database. The results show that semiconductor production capacity is highly geographically concentrated and underscores the importance of distinguishing between different semiconductor types and the limited scope for substitutability of production between fabrication plants. This structural rigidity creates bottlenecks that can amplify the impact of supply disruptions and limit the scope for remedial actions. Enhancing the resilience of the global semiconductor value chain will require further co-operation among like-minded economies, developing better data and exploring opportunities for greater diversification to reduce systemic risks.

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Executive summary

Enhancing the resilience of the semiconductor value chain requires better and more granular data to allow policymakers to identify dependencies, bottlenecks, and help prepare for and react to potential disruptions in value chains.

This paper strengthens the evidence base for semiconductor policy by mapping global semiconductor wafer fabrication capacity by economy, process node density, process technology and business model. It presents the first analysis based on the OECD Semiconductor Production Database which was developed to help inform the OECD's Semiconductor Informal Exchange Network (SIEN). The database compiles and harmonises data on global semiconductor wafer fabrication capacity using commercial datasets and OECD desk research, building on the semiconductor taxonomy developed in 2024 (OECD, 2024[1]) and contributing to better mapping of the global semiconductor value chain (OECD, 2025[2]).

The results show that global wafer fabrication capacity is concentrated in five economies. The People's Republic of China (hereafter "China"), Chinese Taipei, Korea, Japan, and the United States account for nearly 90% of global capacity. Within these economies, capacity is further concentrated in a small number of firms, with the ten largest manufacturing companies accounting for around half of total global production capacity. Upcoming investments are heavily focused on the same major producing economies, suggesting that existing geographic concentration may persist. The distribution of capacity also differs significantly by chip type. For example, advanced logic and commodity memory are dominated by a limited number of highly specialised fabrication plants (fabs), while analog, mature logic, and power semiconductors are typically produced in fabs with mixed manufacturing capabilities.

The analyses in this paper also suggest limited substitutability of production between fabs. For example, a fab optimised for analog or power chips cannot produce advanced logic or memory chips. This structural rigidity creates bottlenecks in specific parts of the value chain, which in turn can amplify the impact of supply disruptions and limit the scope for remedial actions. Enhancing the resilience of the global semiconductor value chain therefore requires a shared understanding of where such bottlenecks lie and how interdependencies can be managed.

This paper shows that further work and better data are needed to adequately answer key policy questions and help policymakers assess dependencies, evaluate diversification strategies, or monitor long-term trends. Improving the quality, granularity, and reliability of semiconductor manufacturing data should thus be in the collective interest of all economies.

Taken together, the findings suggest several implications for future policy work. Co-operation among like-minded economies will be increasingly important to identify and address structural vulnerabilities along the semiconductor value chain, particularly in areas where production is concentrated. Jointly developing more reliable and policy-relevant data – through shared taxonomies, better industry engagement, and complementary national efforts – would help governments base their decisions on better evidence. Finally, exploring opportunities for greater geographic diversification, could contribute to reducing systemic risks and strengthening the overall resilience of the global semiconductor value chain.

1 Introduction

Enhancing the resilience of the semiconductor value chain requires a better understanding of the value chain and supply and demand of different semiconductor types. This enhanced understanding is not possible without better data to allow policymakers to identify bottlenecks, monitor the balance between demand and supply of specific semiconductor types, and manage disruptions in value chains.

A key objective of the OECD's Semiconductor Informal Exchange Network (hereafter the SIEN), convened in June 2023, is to enhance the understanding of semiconductors and help participants move towards more resilient semiconductor value chains (OECD, n.d.[3]).[1] This paper provides an overview of semiconductor front-end production capacity data, building on the semiconductor taxonomy developed by SIEN in 2024 (OECD, 2024[1]) and available data sources.

By focusing on improving transparency and information sharing on front-end manufacturing, this paper provides preliminary information based on the ongoing development of a database underpinning analyses to address the following key questions:

Where are the production facilities for front-end manufacturing located?

Analysis under this workstream sheds light on where various types of semiconductor production capacity are located and provides insights on market structure and thus potential bottlenecks. Moreover, data on the location, capacity and capabilities of front-end manufacturing facilities currently being planned or underway provide insights on whether recent policy efforts (e.g. policy strategies and instruments) are helping diversify semiconductor production and reduce the risks of bottlenecks in the future.

How is the balance between semiconductor demand and supply expected to evolve?

The planned approach is to combine the existing OECD Semiconductor Production Database with forthcoming data on chip demand to help identify segments where installed capacity is growing faster than demand, and therefore creating risks of excess capacity and unnecessary redundancies. While chip demand analysis is not the focus of this paper, future work combining chip capacity with demand data would help policymakers understand which types of chips investments should be encouraged to build a more resilient overall value chain.

By building on front-end manufacturing capacity information and semiconductor demand developments and forecasts, the analysis could help monitor industry cycles. Furthermore, the taxonomy facilitates distinguishing between different semiconductor product types, with the level of product type granularity depending on the availability of both production capacity and demand data at the same level of granularity.

What is the potential for substitution?

It is important to understand if and where production is substitutable between fabrication plants (fabs) in order to analyse gaps and consider policy approached focused on building more resilient value chains.

It would also help better understand the degree of flexibility in front-end chip manufacturing to cope with supply-demand imbalances for certain chips, including in the event of localised disruptions and other shocks.

Comprehending substitutability is critical when planning for managing worst-case disruption scenarios. There are two important dimensions to substitutability:

- **Chip substitutability** describes the ability to replace one specific chip in an end-product with another that performs the same functions with a minimal loss in performance.
- **Fab substitutability** describes the ability to shift the production of one chip to another plant, and if so, the range of chips that can be produced with the same available fab equipment and facilities (or with minor tweaks).

Further work towards addressing these questions will be necessary, including access to more comprehensive and granular data on chip types and process technologies per fab, as part of the development of the OECD semiconductor production database.

The paper is organised as follows: section 2 describes the various data sources, their limitations and how best to address them, section 3 provides the main findings. Finally, section 4 outlines future work in this workstream towards meeting the objectives and answering the policy questions described in section 1.

2 Data

This paper presents the first analysis based on the OECD Semiconductor Production Database. The database is being developed based on the principles elaborated by the Network (OECD, 2024[1]) that are reproduced in Annex A. This database builds on semiconductor production capacity information available from the industry and commercial data providers, augmented by information on other relevant dimensions that could help meet the goals described in section 1. Data are aggregated at the economy level according to different dimensions (e.g., by chip types and/or specific technologies). Data aggregation is required to ensure co-operation with industry and commercial data providers.

Available data sources

This paper focuses on the wafer fabrication stage of the semiconductor value chain, also referred to as front-end manufacturing, which involves the deposition, patterning, and etching of materials to build an integrated circuit on a wafer, based on in the semiconductor taxonomy developed by the OECD (2024[1]). Thus, only front-end fabs have been included in the analyses presented in this paper. The reason for this initial focus is that front-end manufacturing is the most capital-intensive process step of semiconductor production, and it has received a lot of attention from government policies in recent years. The OECD plans to expand the scope of its Semiconductor Production Database to potentially include back-end manufacturing (assembly, test and packaging) and semiconductor manufacturing equipment.

Definitions

The production capacity of a fab is measured in wafer starts per month (WSPM) in 8-inch (8'') equivalents (EQ). WSPM denotes the maximum number of silicon wafers that can be processed in a fab per month. WSPM is independent of the type of chip (e.g., memory, logic, power, or analog), the yield (share of functioning chips per wafer) or the die size. Since fabs process a variety of different wafer diameters, such as 6" (150mm), 8" (200mm) and 12" (300mm), the capacity is typically normalised across wafer diameters by using 8" equivalents. Unless explicitly stated otherwise, all wafer capacity calculations in this paper are given in WSPM in 8'' equivalents.

The database also tracks the status of a fab. Following is the definition of the different status categories for all fabs in the database, in chronological order of the timeline until a fab reaches full production:

- **Planned**: includes fabs whose planning has been mentioned by the company or that have been officially announced. Importantly, this category may include fabs whose construction is planned to start in five or even more years and fabs whose construction will start the next quarter. In the OECD Semiconductor Production Database, 54 out of a total of 1 433 fabs fall under this category.
- **Under construction**: Fabs are categorised as "under construction" if they (facilities, cleanrooms, etc.) are currently being built or if they have been built and are currently being equipped with machines. In this vintage of the OECD Semiconductor Production Database, 53 fabs are currently under construction.
- **Upcoming**: This paper refers to fabs as "upcoming" if they are either "planned" or "under construction".

- **In-production**: Fabs that are currently producing chips, including the "first silicon" run[2], fall under this category. 1 326 fabs out of 1 433 fabs are considered to be in-production for the purposes of analyses presented in this paper.

The analysis includes R&D and pilot lines (145 out of 1 433 fabs), unless explicitly stated otherwise, but excludes fabs involved solely in epitaxy or back-end manufacturing. Micro Electro-Mechanical Systems (MEMS) fabs have been included in the database. While MEMS themselves are not semiconductors many fabs that are capable to produce MEMS are also producing semiconductors. Furthermore, MEMS devices are increasingly integrated with semiconductor devices already during wafer fabrication (Samudrapom Dam, 2023[4]). Analyses on other stages of the semiconductor value chain (design; assembly, testing and packaging) are beyond the scope of this paper.

The analyses in this paper build on highly disaggregated data at the level of the semiconductor fab sourced from two commercial data providers, SEMI (World Fab Watch third quarter 2025) and TechInsights Inc. (300mm Watch September 2024 and Semico Fab Database Summer 2023), and complemented with desk research, described in this section. For further discussion of all data sources please see Annex B.

Data limitations

Currently commercially available datasets about wafer fabrication come with several substantial limitations, requiring data to be interpreted with caution. While addressing the current shortcomings will take considerable time and effort, the findings described in this paper are a first step in an iterative process of constant improvement. Until recently, quantitative data about the global semiconductor industry was collected and prepared *by* the industry *for* the industry, with little consideration for aspects relevant to policymakers, such as geographical diversity of manufacturing. Increased interest by the policy making community in semiconductor data will also support the efforts of the Network towards developing a database with the purpose of addressing policy questions.

Assessing and meaningfully comparing the geographic distribution of wafer fabrication capacities and modes of production faces several challenges. These stem largely from the limitations of currently available data sources as well as general technology trends. As mentioned before, wafer starts per month (WSPM) serves as the primary metric for assessing manufacturing capacity in commercial datasets. Given the reliance on imputed data and estimates from data suppliers, uncertainty analysis (i.e. confidence intervals) is not meaningful as the data reflects industry-informed assumptions rather than direct observations. These educated guesses are standard in industry practices but do not allow for confidence intervals or variability analysis tied to a true underlying value.

The level of aggregation of economies presented in the paper is based on relevant focus on the most concentrated region. Further disaggregation is only possible in the case that the data agreements with the commercial data providers is not violated which is reliant on any disclosure of firm level information.

At the time of writing, WFW and World Fab Forecast (WFF) from SEMI are the most comprehensive, commercially available datasets of front-end fabs (SEMI, 2024[5]). WFW currently lists more than 1 500 fabs with several data points each. Given their comprehensiveness, these commercial databases are currently the primary data sources used for the analysis in this paper. As such, the findings described in this paper rely largely on the accuracy, completeness and correctness of these databases. While they are valuable sources, like any established database evolved over many years, they may have certain limitations.

The following is a brief overview of some of the key constraints of the analysed data and how they affect conclusions drawn from the figures.

When interpreting the figures and analyses presented in this paper, several important data limitations should be taken into account. These limitations arise from inconsistencies across commercial datasets,

missing detail on process technologies and materials, incomplete geographic coverage, and uncertainty surrounding future developments.

- **Process node density:** Inconsistent and incomplete data on process node density (feature size) per fab limit the ability to accurately assess the geographic distribution and substitutability of specific manufacturing capabilities.
- **Process technologies**: Commercial datasets do not consistently capture or disaggregate process technologies (e.g. BCD, RF, analog) per process node and fab, limiting meaningful analysis of technological specialisation and substitutability across fabs.
- **Wafer materials**: While commercial data sources often identify whether fabs can use different wafer materials beyond silicon, such as silicon-carbide (SiC) and gallium-nitride (GaN), they lack information on the production share by material type, hindering assessment of material-specific capacity and capabilities.
- **Geographic coverage**: Data coverage is uneven across economies, especially for China, where limited transparency and delayed reporting result in likely underestimation of actual wafer fabrication capacity.
- **Developments in semiconductor technology**: The growing relevance of advanced packaging and the resulting integration of front- and back-end manufacturing processes blurs traditional capacity metrics and renders assessments solely focused on wafer fabrication capacity increasingly incomplete.
- **Uncertainty of upcoming fabs**: Reported projections for future wafer capacity are uncertain, as planned facilities face delays, downsizing, or cancellation, meaning current datasets represent only a temporary snapshot of an evolving landscape.

Data limitations are discussed in greater detail in Annex B.

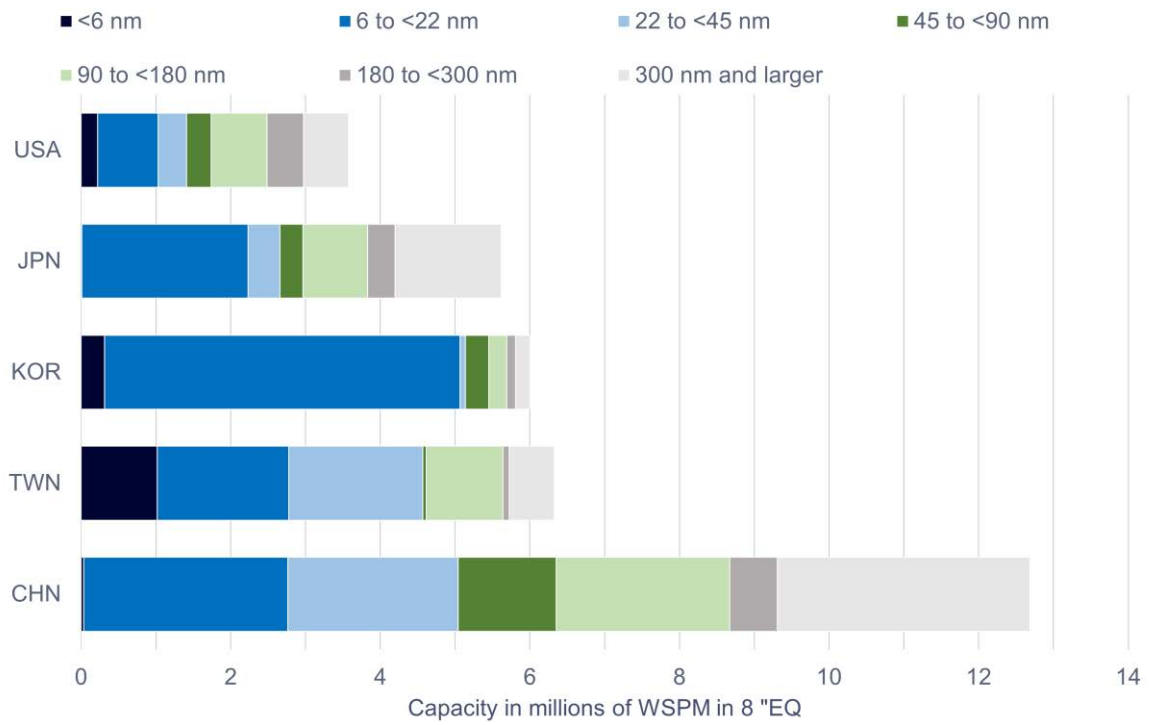
3 Main findings

This section discusses a selection of findings based on the analysis of data from SEMI, TechInsights and desk research. These findings should be interpreted with some caution, in view of the data limitations described above. The results in this section provide insights into the global semiconductor value chain, identify market dynamics and analyse the geographic distribution of production.

Geography of production capacity differs by process node

Figure 1 shows the geographic distribution of in-production wafer capacity (excluding capacity from upcoming fabs) in seven ranges of node densities (see Box 1 for a detailed description of node density ranges) for the five economies with the highest in-production wafer capacity, namely China, Chinese Taipei, Korea, Japan and the United States.[3] As of September 2025, these five economies together account for 87% of global in-production wafer capacity.

Figure 1. Feature size distribution in the five economies with the largest in-production capacity



Note: Please refer to Box 1 for a description of the node density ranges. Only in-production fabs are included here.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Figure 1 illustrates how the mix of process nodes differs by economy. For example, Korea's wafer capacity is highly focused, with almost 80% of its capacity stemming from process nodes between 6 and smaller

than 22nm. This focus is primarily due to SK Hynix and Samsung, Korea's largest semiconductor suppliers, which are heavily invested in memory chip production that relies on constant node upgrades. Since most of today's NAND and DRAM production depends on process nodes ranging from 6nm to <22nm, it explains why Korea's wafer capacity is so focused. In contrast, the United States' wafer capacity is much less focused on specific nodes but rather diversified across a wide range of process nodes.

Box 1. Seven ranges of node densities

The seven ranges used in the analysis are based on several factors, such as coverage of at least 2 full-nodes (see Note), changes in transistor architecture and dependence on advanced lithography equipment. The following is a brief overview of the seven node density ranges.

- **<6nm:** In order to build a 5nm and smaller process node, a fab requires access to extreme ultra-violet (EUV) lithography equipment. 3, 4 and 5nm nodes are currently used to produce cutting-edge mobile system-on-a-chip (SoCs), cloud AI accelerators for machine learning and high-performance processors in consumer devices and servers.
- **6 to <22nm:** This includes two full-nodes (10nm and 14/16nm) and two half-nodes (7nm and 12nm). For process nodes smaller than 22nm, the transistor architecture must be changed from planar transistors to FinFETs, making chip design and the production process more complex and costly (Lapedus, 2018[9]). Today, most of the wafer capacity in this range comes from memory chip fabs producing DRAM and NAND.
- **22nm to <45nm:** This range includes two full-nodes (22nm and 32nm) and two half-nodes (28nm and 40nm). These are very popular process nodes for a variety of chips, such as microcontrollers. A lot of capacity, especially in China, is expected to come online in this range of process nodes over the next few years (Goujon, Kleinhans and Gormley, 2024[10]). Process nodes at 40nm and smaller depend on argon-fluoride immersion lithography (ArFi) equipment.
- **45nm to <90nm:** This range consists of two full-nodes (45nm and 65nm) and two half-nodes (55nm and 80nm). Process nodes in this range are crucial for radio frequency and mixed-signal chips, as well as microcontrollers and embedded (specialty) memory.
- **90nm to <180nm:** This includes the full-nodes 90nm and 130nm and the half-nodes 110nm and 150nm. These are popular process nodes for power management ICs, mixed-signal chips and analog semiconductors.
- **180 to <300nm:** This includes the full-nodes 180nm and 250nm as well as the half-node 220nm. These process technologies are still very relevant for (discrete) power semiconductors and analog chips.

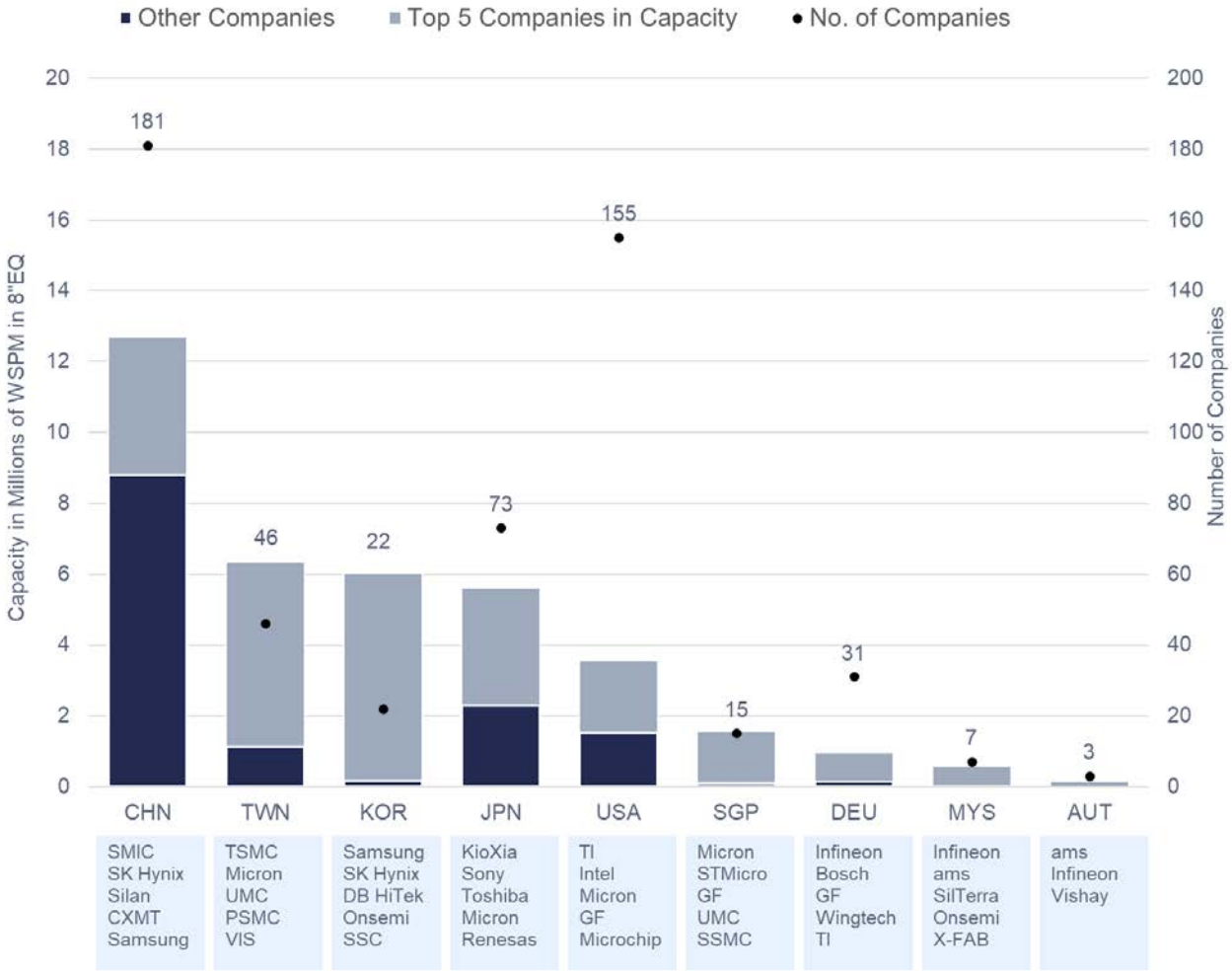
Note: "Full-nodes" represent significant advancements in semiconductor technology with substantial reductions in feature size and major process changes, such as the jump from 45nm to 32nm. "Half-nodes" are intermediate steps that offer smaller, incremental improvements, for example 28nm between 32nm and 22nm. However, this characterisation lost meaning for node densities under 14/16nm, as scaling became more about architectural innovations (e.g., FinFETs) and marketing decisions rather than straightforward geometric reductions.

A large share of production capacity is in a few companies

The ten largest semiconductor companies in terms of production capacity, account for around 50% of WSPM globally. Figure 2 provides insights on the firm distribution in the largest nine semiconductor producing economies. In Japan, 73 companies operate at least one fab, with a combined wafer capacity

of more than five million WSPM. The five companies with the largest wafer capacity in Japan, namely Kioxia, Sony, Toshiba, Micron, and Renesas, account for above three million WSPM (58% of Japan's total). The remaining 42% of Japan's wafer capacity is provided by the other 68 companies operating fabs in the economy. Wafer production is more concentrated in other economies, such as Korea, Chinese Taipei, Singapore and Germany. China is the only economy where the five companies with the highest wafer capacity represent less than half of the economy's total wafer capacity.

Figure 2. Capacity share of the largest five companies by economy



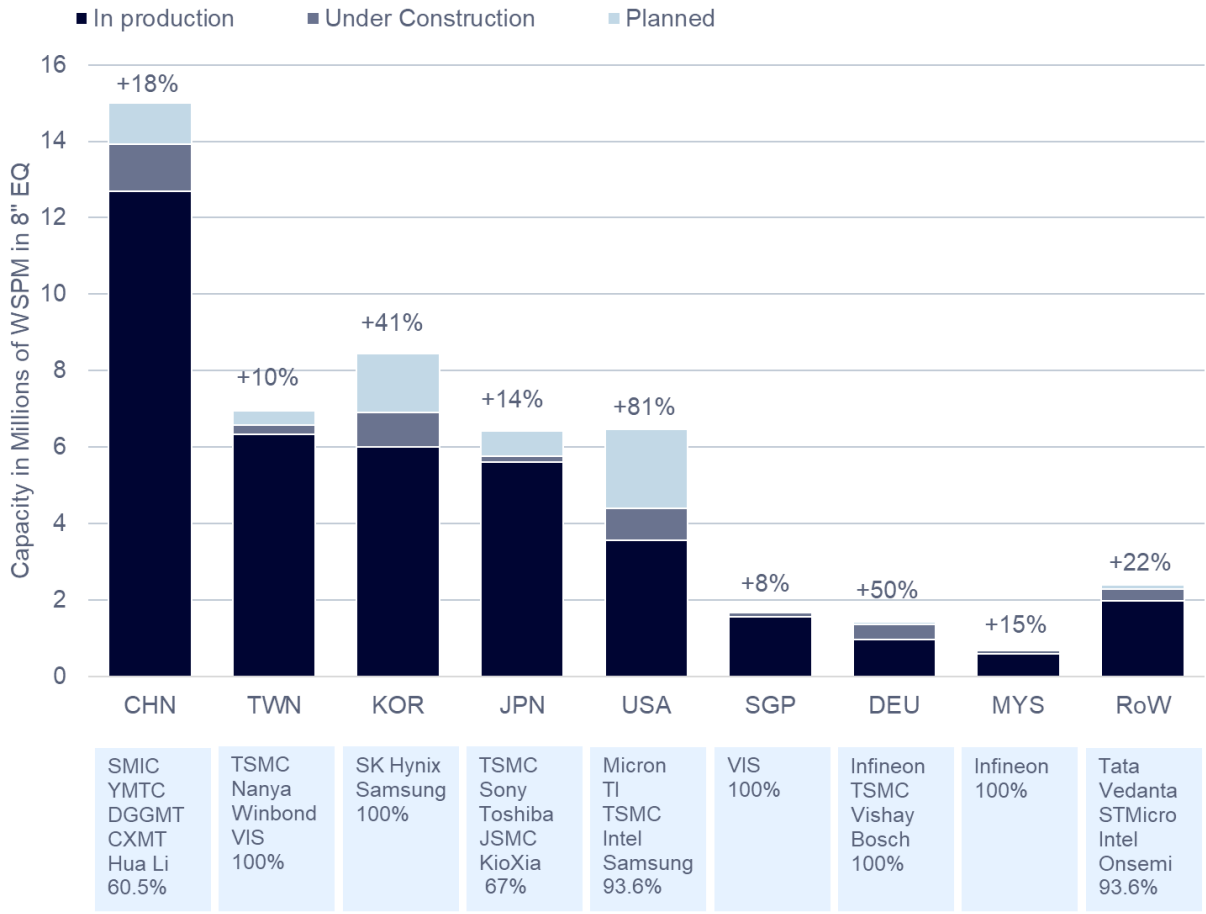
Note: Company share is measured by in-production capacity, in million (M) WSPM in 8'' equivalents. The figure shows data for the nine largest economies in terms of in-production capacity. The light grey bar represents the capacity share of the top five companies (with the largest wafer capacity) and corresponds to the left-hand scale (LHS). The dark grey bar shows the capacity share of the rest of the market and also corresponds to the left-hand scale (LHS). The black dot indicates the number of companies operating fabs in each economy and corresponds to the right-hand scale (RHS). Companies are identified based on the location of the production facilities (not headquarters), thus the same company may appear in multiple economies. The top five companies are listed in descending order of wafer capacity at the bottom of the figure.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Capacity gains of planned and under construction wafer capacity per economy

Figure 3 shows the capacity of planned, under construction and in-production fabs located in the economies with highest capacity. Most of the capacity investments (both green- and brownfield fab projects) are taking place in the largest semiconductor producing economies and is driven mostly by the

largest semiconductor companies operating fabs in the corresponding region. The largest increase in capacity (WSPM) is upcoming in the United States, China, Korea, Chinese Taipei, Japan, Germany, and Singapore. India accounts for the largest share in upcoming capacity among the rest of the world (RoW).

Figure 3. Capacity by economy with top upcoming capacity generating companies



Note: The figure shows total in-production and upcoming capacity, measured in million (M) WSPM in 8" equivalents, for the economies with largest capacity. Rest of World (RoW) aggregates all economies in the dataset not otherwise listed in the chart. The companies accounting for the increase in capacity due to upcoming fabs are included in the light blue box at the bottom, with the percentage of their contribution to the total upcoming capacity is also included in the figure. The percentage on top of each bar denotes the share of upcoming capacity based on the economy's in-production capacity.

Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Figure 3 also shows that more than 60% of planned capacity expansions in each of the economies stems from less than six companies per economy. For example, more than 90% of planned capacity expansions (WSPM) in the United States come from Micron, Texas Instruments (TI), TSMC, Intel, and Samsung, in descending order of upcoming capacity expansions.

While the largest semiconductor producing economies are expected to see their capacity increase, it is important to highlight considerable differences across process technologies and chip type capabilities. Upcoming capacity developments in geographic regions and economies can be analysed through product types and business models to better understand how the chip landscape is forecasted to evolve over the next years.

Wafer capacity by chip type

Node density (nanometres) alone is not a meaningful metric for assessing the geographic distribution of wafer capacity. Process technologies (chip types) and business models (i.e. fabs operated by foundries or IDMs) have to be taken into account as well. When examining the geographic distribution of wafer capacity based on the types of chips a fab is capable to produce, substantial differences emerge.

Importantly, in the following Figure 4 and Figure 5, fabs may be counted multiple times because each panel reflects their distinct “capabilities”. For example, a single fab from Tower Semiconductor might offer Bipolar-CMOS-DMOS (BCD) process technology for power semiconductors, radio frequency silicon-on-insulator (RF SOI) for analog RF chips, and non-volatile specialty memory (NVM). Thus, this fab would be included in the wafer capacity totals for three different categories – analog, power/discrete and specialty memory – in a given economy. Furthermore, within each category, the fab’s entire wafer capacity would be counted. For instance, if a fab with 40 thousand WSPM offers process technologies for both analog semiconductors and specialty memory chips, the fab’s total capacity will be counted twice – once for analog and once for specialty memory (see section 2 for explanation).

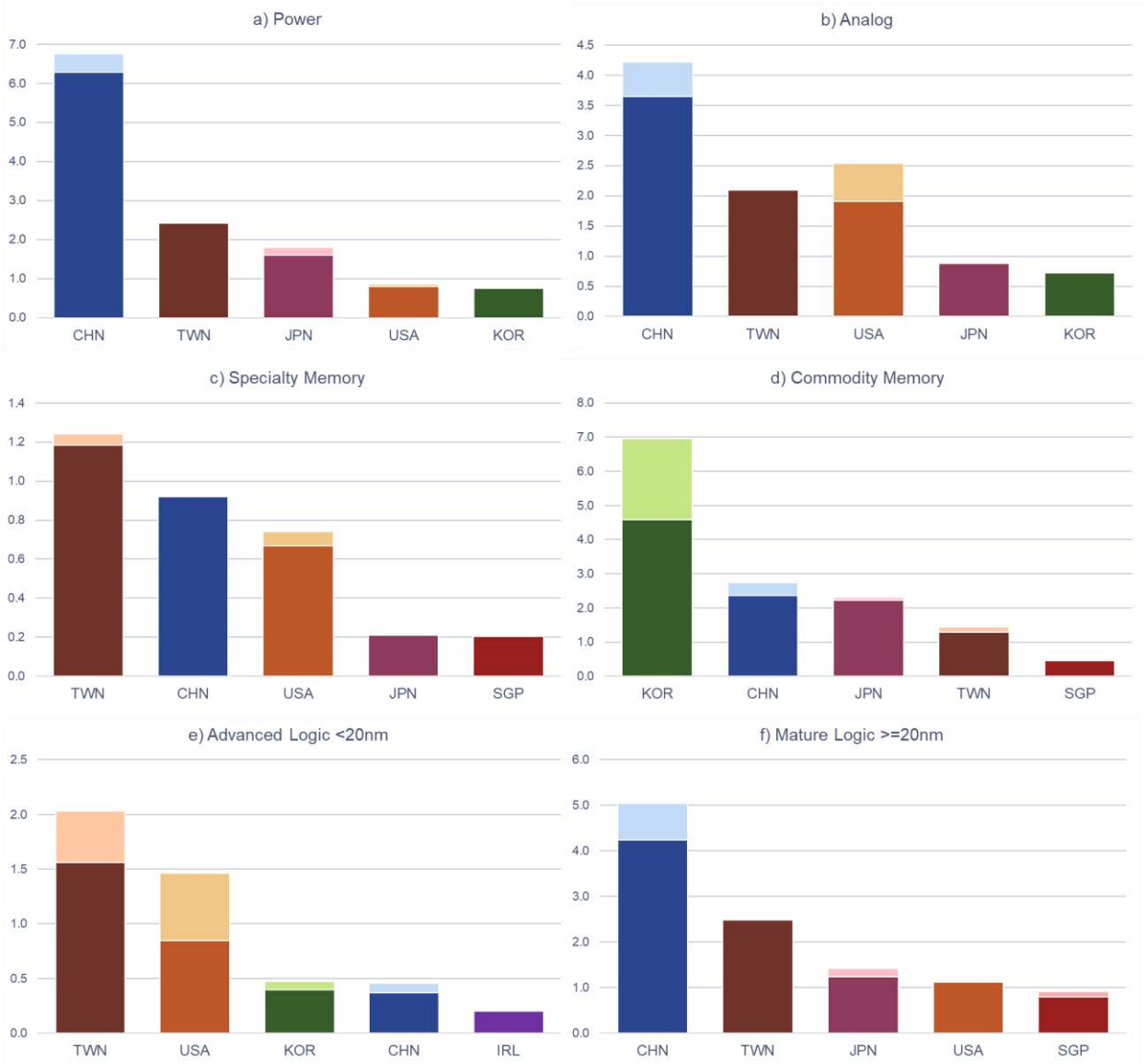
Thus, Figure 4 and Figure 5 have several limitations, and the data should be interpreted with caution. As indicated earlier, this analysis heavily relies on the completeness, correctness and accuracy of the underlying data regarding process technologies. The information presented in Figure 4 and Figure 6 can entail important biases if available data do not accurately and comprehensively represent the variety of process technologies offered in a specific fab – see 4Annex B for a discussion of process technology offerings per technology node. Given the variety of process technologies per fab and process node, data are likely incomplete. Therefore, the geographic distribution of capacity shares per chip type should be viewed as rough indicators rather than definitive assessments.

Select economies with highest wafer capacity per chip type

The six panels in Figure 4 show the total wafer capacity (WSPM) for six different types of chips, namely power and discrete; analog; mature logic (>=20nm); advanced logic (<20nm); commodity memory (DRAM and NAND)[4]; and specialty memory.[5] These distinctions between types of logic chips and types of memory chips are used solely for the purpose of the analysis in this paper. The Network is welcome to provide guidance on how best to distinguish these types of chips.[6]

For each of these types of chips, Figure 4 shows the five economies with the highest total wafer capacity, based on the geographical location of the fabs rather than the headquarter location of the companies. The darker shade denotes the in-production capacity, the lighter shade shows upcoming (under construction and planned) capacity.

Figure 4. Five largest economies by wafer capacity per chip type



Note: Each panel shows an economy's total wafer capacity for different types of chips (based on different process technologies), measured in million (M) WSPM in 8'' equivalents. The dark shade corresponds to the capacity accounted for by the in-production fabs while the light shade reflects upcoming fabs in each of the economy.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

China and Chinese Taipei are the only economies that feature in the top five largest chip producers across all six chip types. The United States and Japan follow, appearing in the top five for all chip types except commodity memory and advanced logic, respectively. However, if all upcoming commodity memory fabs in the United States come to fruition, the United States would also rank among the top five economies for all six chip types. Korea and Singapore are also important players in several types of chips.

Production capacity for **power/discrete chips** is led by China, with 6.28 million WSPM, followed by Chinese Taipei with 2.42million WSPM and Japan with 1.60 million WSPM. China also has the largest total capacity for analog chips, with 3.64 million WSPM, followed closely by Chinese Taipei (2.09 million WSPM) and the United States (1.90 million WSPM).

In logic chip types, in-production capacity in **mature nodes** is led by China (4.23 million WSPM), followed by Chinese Taipei (2.48 million WSPM) and Japan (1.24 million WSPM). For **advanced logic chips**, Chinese Taipei leads (1.55 million WSPM), followed by the United States (0.84 million WSPM) and China (0.39 million WSPM).

In memory chip types, Korea has the highest wafer in-production capacity for **commodity memory** (DRAM and NAND) by a significant margin, which is not the case for specialty memory, such as NOR Flash. Korea leads on commodity memory capacity (4.58 million WSPM), followed by China (2.37 million WSPM and Japan (2.21 million WSPM). In **specialty memory**, Chinese Taipei (1.18 million WSPM) is closely followed by China (0.92 million WSPM) and the United States (0.67 million WSPM).

Select economies with highest upcoming capacity expansions per chip type

Upcoming expansions of capacity are not evenly distributed across chip types and economies. Figure 5 provides a more granular view of upcoming wafer capacity by chip type and economy to better assess differences between economies. For certain chip types, such as commodity memory and advanced logic, the upcoming capacity is mainly in the economies with largest in-production capacity. Meanwhile, the upcoming capacity for power, analog, mature logic, and specialty memory chips is taking place in several other economies as well, even if it would not amount to significant changes in the list of top five economies with highest capacity. Substantial capacity additions in all six different chip type categories can only be observed in China and the United States. Capacity developments in other economies are much more focused on specific types of chips.

Upcoming capacity for **power chips** is taking place mostly in China, with additional 0.48 million WSPM, followed by Germany (0.32 million WSPM) and Japan (0.19 million WSPM). If these upcoming capacity expansions come into production, the top five economies with largest power chip capacity would be China, Chinese Taipei, Japan, Germany, and the United States respectively.

For **analog chips**, the upcoming capacity is led by the United States with an additional 0.64 million WSPM, followed by China (0.57 million WSPM) and then Germany (0.27 million WSPM). Should the production capacity underway materialise, the leading economies with the highest analog chip production capacity would be China, the United States, Chinese Taipei, Japan and Germany.

For **mature logic chips**, the upcoming capacity would be largely focused on China. Companies in China would build out more than three times the combined upcoming wafer capacity (0.81 million WSPM) of the other top six economies in Figure 5: capacity expansions from Japan (0.18 million WSPM) and Germany (0.09 million WSPM), amount to 0.27 million WSPM. Based on the data analysed, Chinese Taipei, United States, and Korea do not have upcoming mature logic wafer capacity. This would correspond to China, Chinese Taipei, Japan, the United States, and Singapore being the leading economies with the highest capacity in mature logic chips.

For **advanced logic chips**, the increase in capacity would be mainly in the United States with additional 0.62 million WSPM, followed closely by Chinese Taipei (0.47 million WSPM). Thus, the United States and Chinese Taipei together would build out twice the capacity (1.09 million WSPM) of the of the other top-six economies in Figure 5 combined (0.52M WSPM). This would lead to a list of the largest top five advanced logic chip producing economies led by Chinese Taipei, the United States, Korea, China, and Japan.

In **commodity memory chips**, the highest increase in capacity would take place in Korea with additional 2.36 million WSPM, more commodity memory capacity than the other five economies combined. USA (1.71 million WSPM) and China (0.38 million WSPM) are poised to build out the second and third most commodity memory capacity, respectively. Should these capacity developments materialise, Korea, China, Japan, Chinese Taipei, and the United States would be the largest commodity memory chip producing economies.

For **specialty memory chips**, the upcoming capacity is focused on United States (0.074 million WSPM) and Chinese Taipei (0.057 million WSPM). The top five economies with the highest specialty memory capacity would be Chinese Taipei, China, the United States, Japan, and Singapore.

Figure 5. Capacity expansion by product type for the top six capacity expanding economies



Note: The figure displays upcoming capacity, measured in million (M) WSPM in 8'' equivalents, disaggregated by chip types for the six economies with the highest projected upcoming capacity values in descending order.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Fabs with mixed manufacturing capabilities (process technologies)

Most fabs can manufacture more than one type of chip, which makes it challenging to analyse the geographic distribution of wafer capacity based on the types of chips a fab is capable to produce. Moreover, uneven coverage of the dataset means that there are fabs that are not attributed to any of the chip type categories. Figure 6 illustrates the heterogeneous nature of chip manufacturing, with the exclusion of 375 unattributed fabs in the OECD production database. The figure highlights the overlaps in chip types for fabs in addition to the exclusive fabs of the particular chip type categorisation. As fabs have heterogeneous capabilities, one fab can produce more than one type of chip. However, there are some fabs only producing one chip type.

- **Specialty memory**: NOR Flash and other types of specialty memory are produced in low volumes, compared to the other five types of chip categories in the dataset (see Figure 4). Out of the 72 fabs

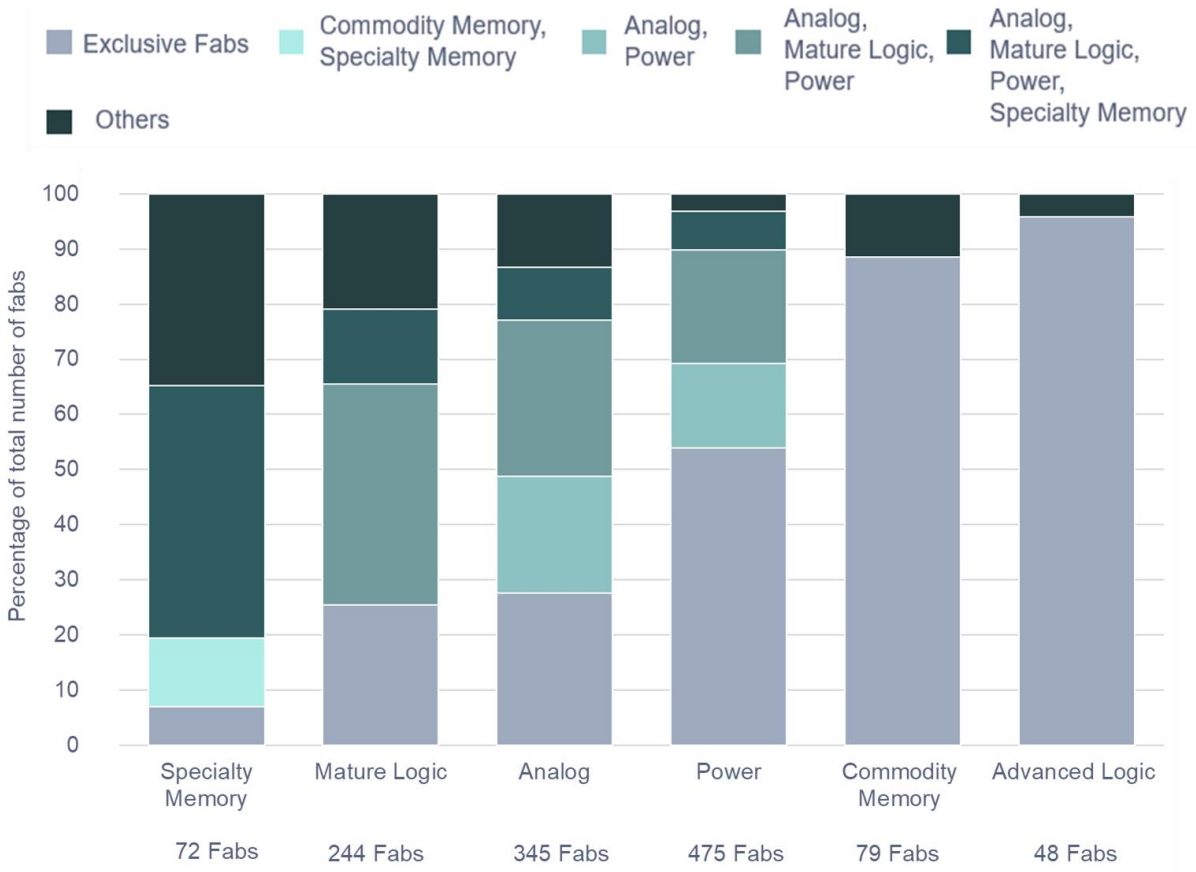
in the dataset capable to produce specialty memory only one fab produces just specialty memory chips. Around 12% of fabs capable to produce specialty memory also produce commodity memory. More than 80% of fabs capable to manufacture specialty memory also manufacture analog, power or mature logic chips.

- **Mature logic**: Out of the 244 fabs in the dataset capable to produce mature logic chips (>=20nm) around 25% are exclusively producing mature logic chips and no other type. However, roughly 40% of the fabs is also capable to produce analog and power semiconductors. Around 75% of fabs in the dataset are producing mature logic chips in combination with any other type.
- **Analog**: Similar to fabs producing mature logic chips, only 27% of the 345 analog fabs are exclusively producing analog chips. Most fabs capable to manufacture analog chips can also produce power semiconductors or mature logic chips.
- **Power**: Out of the fabs in the dataset, 475 can produce power semiconductors. More than half of those 475 fabs (256 fabs) are only producing power semiconductors. This is in contrast to specialty memory, mature logic and analog chips that are typically produced by "mixed capabilities" fabs.
- **Commodity memory:** Of the 79 fabs in the dataset capable of producing commodity memory chips, such as NAND and DRAM, more than 88% are not producing any other type of chip. Typically, commodity memory chips are not produced by fabs with mixed capabilities.
- **Advanced logic:** Almost all (96%) of the 48 fabs capable of producing advanced logic chips (<20nm) in the dataset are exclusively producing advanced logic chip.

In conclusion, analog, mature logic and specialty memory chips are predominantly produced by fabs that offer a variety of process technologies, enabling them to produce these different types of chips. However, for power semiconductors, exclusive manufacturing is much more common – around half of power semiconductor fabs in the dataset only produce this type of chips. Commodity memory fabs and advanced logic fabs are highly specialised and are generally not capable of producing other types of chips.

This prevalence of "mixed-capability" fabs, especially in mature logic and analog chips, also poses challenges in assessing capacity across different markets since these fabs are not limited to manufacturing a specific type of chip – in contrast to advanced logic and commodity memory fabs. They often seem to offer a broad variety of process technologies, allowing them to shift production and product mix depending on how the market develops. This is one of the reasons making it challenging to assess whether or not excess wafer capacity exists in, for example, mature logic chip production.

Figure 6. Chip type capabilities shared across fabs



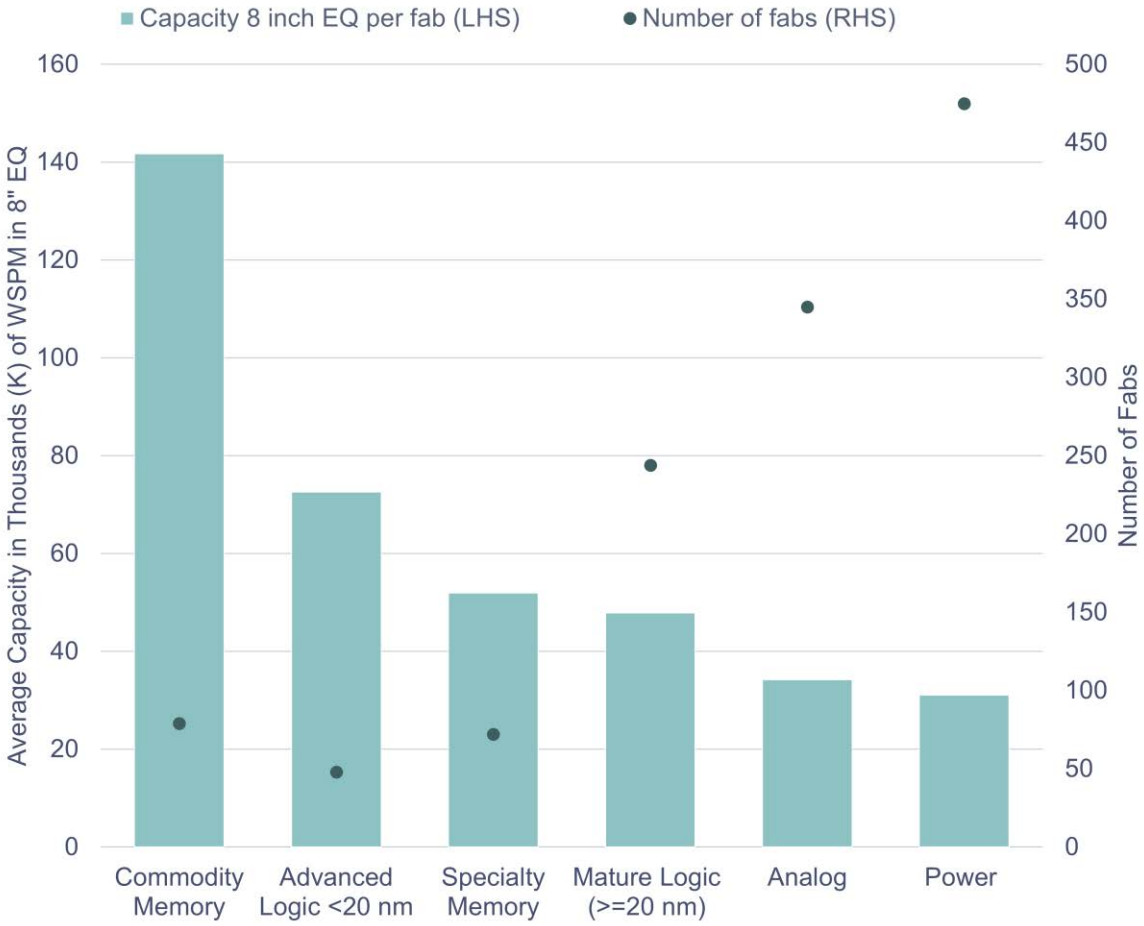
Note: Grey highlights the percentage of total fabs in one chip type category that exclusively produce that particular chip type. Most commonly occurring overlaps among the chip types are presented with the various shades of green where the darkest shade represents the share of fabs producing a mix of chip types not listed otherwise. R&D and pilot lines are excluded. 375 fabs are unattributed to any chip type category and thus also omitted, 810 unique in-production fabs included in the figure.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Average fab size per chip type

Lastly, assessing wafer capacity by chip type also provides insights into the different sizes of fabs. Figure 7 reveals that, on average, fabs substantially differ in size (measured in WSPM in 8" equivalents) depending on the type of chip they are producing. While the average fab size for power, analog and mature logic chips ranges between 30 thousand to 50 thousand WSPM, advanced logic and especially commodity memory fabs are substantially larger in their manufacturing volumes. For example, several of the upcoming 12-inch commodity memory fabs from Samsung, SK Hynix and Micron are planned with a wafer capacity of 150 to 200 thousand WSPM, corresponding to 330 to 450 thousand WSPM in 8" equivalents.

Given increasing capital expenditures for commodity memory and advanced logic fabs, mainly driven by more complex and thus more expensive manufacturing equipment, the average size of fabs for these chip types is also increasing to benefit from economies of scale. This is illustrated by the much larger average size of advanced logic fabs (72 thousand WSPM) compared to mature logic fabs (47 thousand WSPM).

Figure 7. Average fab size by chip type



Note: The figure shows average in-production capacity per fab for six chip types measured in thousands (K) WSPM in 8'' equivalents on the left-hand scale (LHS) and total number of fabs for each chip type category on the right-hand scale (RHS). The share of capacity for each of the product type is not mutually exclusive as one fab can fabricate more than one type of chip type. R&D and pilot lines are excluded. 375 fabs are unattributed to any chip type category and thus also omitted, 810 unique in-production fabs included in the figure.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Ownership and wafer capacity

While the previous section focused on various insights in relation to the process technologies and corresponding chip types fabs offer, this section looks at business models and company ownership. Figure 8 details the wafer fabrication capacity in each economy, distinguishing between fabs owned by domestic and foreign companies. Data clearly show that, in the five economies with the largest wafer capacity, most of the production capacity belongs to domestically owned companies.

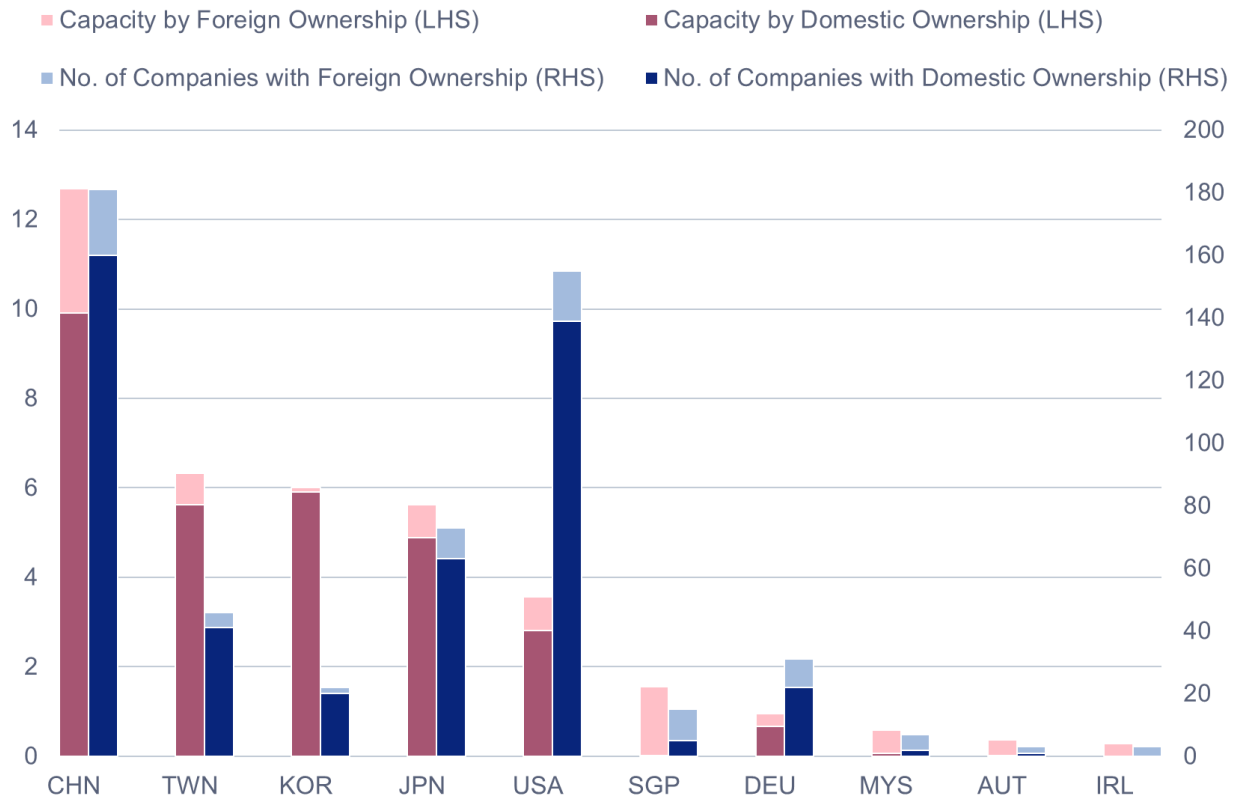
While the global semiconductor value chain is transnationally highly interdependent, most of the capacity is in companies operating in their own jurisdiction (Baisakova and Kleinhans, 2020[11]). One possible explanation is the familiarity with the local regulatory framework and ecosystem: building a fab is capital-intensive, requires access to highly specialised construction companies, technology and infrastructure suppliers while navigating complex regulatory environments (European Commission, 2022[12]).

Nevertheless, recent developments suggest that the share of foreign owned companies operating fabs may increase in some of the largest semiconductor producing economies. For example, TSMC (Chinese

Taipei) and Samsung (Korea) are expanding their wafer capacity in the United States and Japan (Moriyasu, Fang-Ting and Li, 2024[13]; Moriyasu et al., 2024[14]). In many smaller semiconductor producing economies, such as Singapore, Malaysia, Austria and Ireland, most wafer capacity has been developed by foreign companies. Singapore and Malaysia have been notably successful in attracting fab investments from foreign owned companies.

Importantly, ownership structures in the semiconductor industry can be rather complicated and are not meticulously tracked by commercial datasets. As one example, STMicroelectronics is incorporated in the Netherlands, is legally headquartered in Switzerland and has fabs in Singapore, France and Italy (in descending order of wafer capacity as of September 2024). However, most of the commercial datasets categorise it as a Swiss or Italian company. The OECD plans to improve the tracking of ownership, headquarter and production locations in the Semiconductor Production Database going forward.

Figure 8. Share of in-production capacity by domestic and foreign owned companies



Note: The left-hand scale (LHS) measures an economy's in-production capacity, measured in million (M) WSPM in 8'' equivalents – the pink bar shows wafer capacity controlled by domestically owned (dark pink) and foreign owned (light pink) companies. The right-hand scale (RHS) measures the number of domestically (dark blue) versus foreign owned (light blue) companies operating fabs (at least one) in an economy.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

Wafer capacity by business model

Business models for semiconductor fabrication are also evolving. While a fab is normally operated by either a pure-play foundry or an IDM, some IDMs (e.g. Intel, Samsung) adapted their business model to also offer foundry services in some or all their fabs (known as "IDM-foundries"). Business model is therefore another important consideration, when analysing the geographic distribution of wafer capacity. The

analysis in this section distinguishes between i) IDM capacity, ii) pure-play foundry capacity and iii) IDM-foundry capacity.[7]

Foundry capacity allows for supplying the entire market. Any company designing a chip can use the fabs of a foundry, whereas the wafer capacity of a fab operated by an IDM only serves the IDM's internal business consumption.[8] With the rise of application specific chips designed and optimised for specific purposes, an increasing number of fabless and system companies are designing their own chips, thus driving increasing demand for foundry capacity (Thompson and Spanuth, 2021[15]). It is therefore crucial to assess the balance between IDM capacity on the one hand and foundry or IDM-foundry on the other for different geographies.

Figure 9 presents wafer capacity data (in WSPM 8-inch EQ) from the top 50 semiconductor manufacturing companies, covering 82% of the total global wafer capacity. The categorisation of IDM, IDM-foundry, and pure-play foundry is based on the operational model of each fab, and the wafer capacity is assigned to the economy where each fab is physically situated, not where the companies are headquartered. The Secretariat will continue efforts to enhance data quality and accuracy, including on the categorisation according to business model.

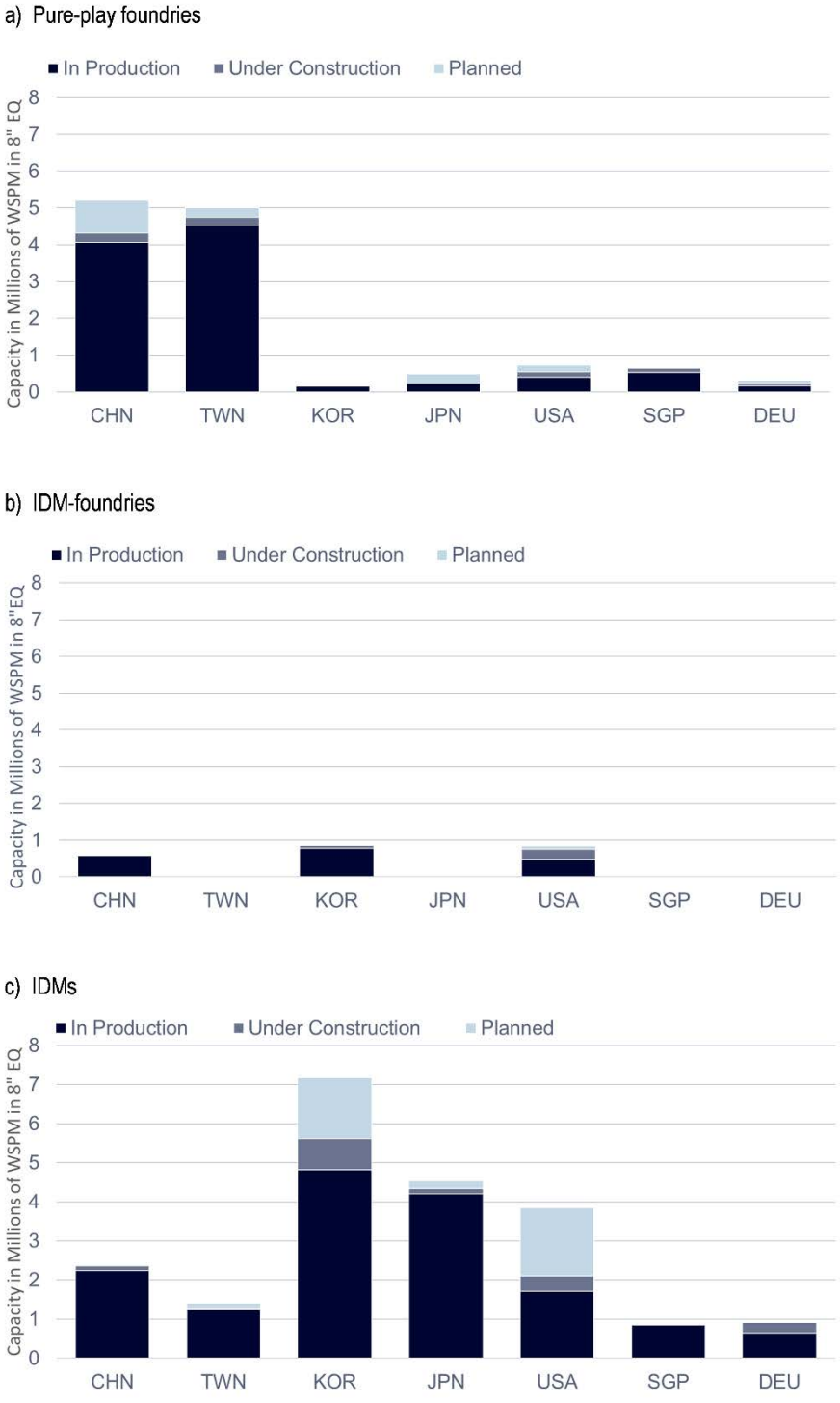
Figure 9 shows that most of the global foundry capacity is in China and Chinese Taipei. China and Chinese Taipei are also the only economies where more than 50% of the domestic wafer capacity comes from foundries and not IDMs.

The relative share of IDM to foundry capacity (including IDM-foundries) per economy is likely to shift, especially for economies such as Japan and the United States, in view of planned wafer capacity by IDM- and pure-play foundries. However, taking the United States as an example, even with the increased investments by IDM-foundries, more than half of the total capacity within the United States would remain IDM capacity.

The stark difference in business models by economy can be explained both historically and by chip types. For example, commodity memory chips are solely produced by IDMs which explains why Korea has very little pure-play foundry capacity – the two largest semiconductor companies in Korea, Samsung and SK Hynix, are IDMs focused on commodity memory and logic chips. Another example would be Japan's semiconductor ecosystem that is also dominated by IDMs.

Importantly, Figure 9 is not exhaustive since the underlying data consist only of the 50 companies with highest wafer capacity, accounting for 82% of global wafer capacity. For example, several Chinese IDMs are planning to expand manufacturing capacity, but they are not included in the dataset. In future updates of the dataset the number of companies with verified business models (IDM, IDM-Foundry, Pure-Play Foundry) will be steadily expanded to improve the quality of the data.

Figure 9. Geographic distribution of wafer capacity by business model based on top 50 companies



Note: The data are based on the top 50 semiconductor manufacturing companies globally, representing 82% of total global wafer capacity. The figure shows in-production and upcoming capacity, measured in million (M) WSPM in 8'' equivalents for seven economies, categorised by the business model of the fab. The business models included are pure-play foundries (panel a), foundries operated by IDMs (panel b), and IDM (panel c). The shade ranges from darkest to lightest colour indicates the status of the fabs from currently in-production, under construction and planned, respectively.
Source: *OECD Semiconductor Production Database* based on augmented SEMI (2025[6]), Semico Research (2024[7]), TechInsights (2024[8]).

4 Conclusion and future work

This paper provides an overview of semiconductor front-end production capacity data, building on the OECD semiconductor taxonomy (OECD, 2024[1]) and the ongoing development of the OECD Semiconductor Production Database to help policymakers understand the complex geography of semiconductor front-end manufacturing and inform actions towards more resilient global semiconductor value chains. It contributes to broader efforts to map the global semiconductor value chain (OECD, 2025[2]).

The analysis highlights the importance of considering at least three key characteristics of fabs - node density, process technology and business models - to understand the geographic distribution of current and upcoming semiconductor manufacturing capacity and interdependencies between companies and economies.

Global wafer fabrication capacity is concentrated: five economies, namely China, Chinese Taipei, Korea, Japan, and the United States, together account for nearly 90% of global capacity. Within these economies, capacity is further concentrated in a small number of firms, with the ten largest manufacturers accounting for around half of total global production capacity. Upcoming investments are heavily focused on the same major producing economies, suggesting that existing geographic concentration may persist.

The distribution of capacity also differs significantly by process technology and chip type. For example, advanced logic and commodity memory are dominated by a limited number of highly specialised fabs, while analog, mature logic, and power semiconductors are typically produced in fabs with mixed manufacturing capabilities.

Data also suggest limited substitutability of production between fabs. For example, a fab optimised for analog or power chips cannot produce logic or memory chips. Even within the same process node, different process technologies are not interchangeable. This structural rigidity creates bottlenecks in specific parts of the value chain, which in turn can amplify the impact of supply disruptions and limit the scope for remedial actions. Enhancing the resilience of the global semiconductor value chain therefore requires a shared understanding of where such bottlenecks lie and how interdependencies can be managed.

However, the analysis also reveals that current commercial datasets are not sufficient for answering key policy questions. They do not adequately capture technological detail, they vary in completeness and consistency across regions and they are currently not designed for policy use. This currently limits the ability of policymakers to assess dependencies, evaluate diversification strategies, or monitor long-term trends. Improving the quality, granularity, and reliability of semiconductor manufacturing data should thus be in the collective interest of all economies.

Taken together, the findings suggest several implications for future policy work. Co-operation among like-minded economies will be increasingly important to identify and address structural vulnerabilities along the semiconductor value chain, particularly in areas where production is concentrated. Jointly developing more reliable and policy-relevant data – through shared taxonomies, better industry engagement, and complementary national efforts – would help governments base their decisions on better evidence. Finally, exploring opportunities for greater geographic diversification, could contribute to reducing systemic risks and strengthening the overall resilience of the global semiconductor value chain.

The OECD Secretariat will continue analytical work to support semiconductor policymaking, including, for example, on combining available production capacity data with information on chip demand. This work

aims to assess whether risks of excess capacity and unnecessary redundancies can be identified, and to understand which chip types should be prioritised for investments to ensure a more resilient value chain.

Additional work will involve extending the data upstream and downstream from front-end manufacturing. The downstream extension, including back-end manufacturing (assembly, test and packaging), could provide information on how different types of chips depend on a combination of front-end and back-end manufacturing technologies. The upstream extension, including information about semiconductor manufacturing equipment and subsequently chemicals and other material inputs, would potentially provide insights into how different types of front-end manufacturing depend on specific types of manufacturing equipment.

Endnotes

[1] For additional information about the OECD Semiconductor Informal Exchange Network, please visit: https://www.oecd.org/en/networks/semiconductor-informal-exchange-network.html.

[2] “First silicon" refers to the initial wafers processed through the full production flow to validate equipment, processes, and integration, also commonly called a "fab qualification run" or "pilot run".

[3] When treated as a single economy for the purposes of this assessment, the EU27 would hold the sixth position worldwide in wafer manufacturing capacity, with approximately 2 million WSPM, just above that of Singapore.

[4] DRAM (Dynamic Random-Access Memory) is a type of volatile memory is commonly used in computers and servers for temporary data storage, whereas NAND Flash is a non-volatile memory used for long-term data storage due to its ability to retain data without power — see the OECD semiconductor taxonomy (OECD, 2024[1]) for additional details.

[5] Specialty memory refers to memory types designed for specific applications and use cases, often prioritising attributes like durability, speed, or power efficiency. Unlike standard memory like DRAM or NAND, specialty memory includes technologies such as NOR Flash, MRAM, and ReRAM, which are tailored for tasks in industries like automotive, industrial, and IoT, where specific performance characteristics are essential.

[6] The discussions of the Semiconductor Informal Exchange Network on the categorization of chip types will help inform a future revision of the OECD taxonomy (OECD, 2024[1]).

[7] For example, Samsung, as an IDM-foundry, operates memory fabs (DRAM and NAND) as an IDM in Korea and China. Samsung also operates fabs as an IDM-foundry for contract manufacturing in Korea and the United States.

[8] Some types of chips, such as commodity memory, are only produced in an IDM setting since competition heavily relies on economies of scale.

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Annex A. Principles of a Semiconductor Production Database

This Annex presents the set of principles for developing the semiconductors production database, based on the discussions of the Semiconductor Informal Exchange Network on a semiconductor taxonomy (OECD, 2024[1]). These principles reflect a prioritisation to help attain the objectives outlined in Section 1.

Trust. The taxonomy, any data sharing activities and resulting databases developed in the context of the Network should build on co-operation and trust among governments and stakeholders in the semiconductor industry to ensure that data and related analyses meet shared goals. Respecting constraints associated with sharing granular industry data would be particularly important to building trust with stakeholders.

Availability. The taxonomy should reflect available data. Areas for which data are not available are not included. Areas for which data availability is very limited are identified as such and, as in any database, there may be numerous missing values. Data analyses are only possible insofar as data are available. Any data collection efforts should complement and rely on, rather than duplicate, existing data efforts by governments, industry and other stakeholders.

Comparability. Comparison across classifications used in different economies and jurisdictions should be as simple and easy as possible. A taxonomy should also be consistent over time, thus allowing for analysis that can build on historical data.

Tractability. Semiconductor manufacturing processes are highly complex. The taxonomy should strive to simplify as much as possible to obtain a tractable taxonomy, noting that this may lead to nuanced differences in semiconductor manufacturing processes and products being obscured.

Adaptability. The taxonomy should be able to adapt as semiconductor production processes and products evolve, keeping abreast of innovation and limiting the needs for revisions of the taxonomy, but without prejudice to the above principle of comparability (through time).[1]

Granularity. Allow for the highest level of disaggregation possible with available data. More granular data offers additional information and allows for more detailed analysis, noting that a higher level of granularity might increase complexity. Input from the Network during its second meeting on 19-20 September 2023 highlighted the importance of ensuring that policymakers can draw meaningful insights from the data. This suggests that **availability** and **tractability** should be prioritised over granularity. Granularity should not hinder efforts to build **trust** and ensure the longstanding engagement of stakeholders. Granularity can be increased in further versions of the taxonomy, if needed and upon availability of the corresponding data and sufficient resources.

These principles may be reviewed in the future to ensure that they remain applicable in practice and well-suited for the objectives of the SIEN – e.g. granularity might become more relevant with the development of the resulting semiconductor production database. The relative importance of the different principles may also change depending on how critical different chips are – e.g. when analysing more critical/vulnerable and non-substitutable chips it might be worth considering putting more weight on granularity and adaptability.

Annex B. Data sources and limitations

Data sources

The analyses in this paper build on highly disaggregated data at the level of the semiconductor fab sourced from two commercial data providers, SEMI and TechInsights Inc., and complemented with desk research.

SEMI's World Fab Watch

SEMI is a global industry association that also offers a variety of commercial datasets, mainly focusing on the supplier markets, such as semiconductor manufacturing equipment and materials. Their World Fab Watch (WFW) is a quarterly updated dataset of all wafer fabrication facilities globally (SEMI, 2025[6]). WFW provides comprehensive information about the location, volume, technical capabilities and planned expansions (capital expenditures) per fab. The analyses presented here are based on SEMI's 2025 third quarter reporting of the WFW dataset.

TechInsights' 300mm Watch and Semico Fab Database

The 300mm Watch from TechInsights provides similar information to SEMI's WFW but focuses only on fabs that process 300mm wafers (TechInsights, 2024[8]). While today's advanced logic and memory fabs are all using 300mm wafers, compound semiconductors, such as gallium-nitride (GaN) and silicon-carbide (SiC) are just transitioning from 150mm to 200mm wafer fabrication. Therefore, TechInsights' data need to be complemented with other datasets to get a more comprehensive picture of front-end production capacity. The September 2024 version of the 300mm Watch dataset is the basis for the analyses presented in this paper.

TechInsights also provides the Semico Fab Database, which is the third dataset contributing to the Network's overview of the global production landscape (Semico Research, 2024[7]). The production data analyses presented in this paper use the Semico Fab Database Update Summary 2023 as one of the fab data sources.

Desk research

Neither of the datasets are fully complete, correct and accurate. With many tracked fabs globally and a rapidly evolving ecosystem, missing information, coverage biases and inaccuracies can be expected. Moreover, existing datasets are usually designed for other purposes than policy analyses and hence might lack information relevant for policymakers. Therefore, the OECD also carried out desk research to verify, correct and complement the data available from the commercial data sources described above. Desk research includes financial reporting from companies, expert interviews, exchanges with the Network participants and media reporting from varied sources.

Data limitations

When interpreting the figures and analysis presented in this paper, several important data limitations should be taken into account, as they affect the accuracy and comparability of the underlying wafer

fabrication data. These limitations arise from inconsistencies across commercial datasets, missing detail on process technologies and materials, incomplete geographic coverage, and uncertainty surrounding future developments.

Process nodes

The feature size[2], generally measured in nanometres (nm), is a very rough proxy for the technological advancement or maturity of the chips that a fab can produce. This is therefore an important variable to consider when assessing capability, in particular of logic fabs.

Nevertheless, these types of data present some challenges. For example, when assessing only geometries (density) of process nodes – such as 7nm, 14nm or 28nm process nodes – it becomes evident that there are recurring discrepancies between public information provided by companies collected through desk research and what is reported in the commercial datasets described above. For example, at the time of writing, Samsung's foundry business lists the following logic process nodes on their website: 3, 4, 5, 7, 8, 14, 28, 70, 90, 130 and 180nm (Samsung, n.d.[16]). However, WFW lists only the following in-production logic process nodes for Samsung's foundry business: 3, 8, 9, 11, and 28. For comparison, TechInsights' 300mm Watch lists in-production Samsung's foundry offerings as 3, 5, 10, 14, 20, 28, 40 and 65nm. GlobalFoundries is another example: The company states they offer 12, 22, 28, 40, 45, 55, 90 and 130nm processes (GlobalFoundries Inc., 2022[17]). WFW does not include the 28, 55 and 130 processes and adds 9nm and 180nm process nodes for GlobalFoundries.

These discrepancies in tracking wafer capacity per process node density extend to individual fabs as well. While SEMI only lists one feature size per fab (often, but not always, the densest feature size a fab is capable of), TechInsights provides the range of feature sizes with rough capacity shares. For instance, the same fab might be recorded as 40nm in SEMI but listed with 40, 65, 90 and 130nm in TechInsights, with estimates of the respective wafer capacity shares. Thus, given the characteristics of the commercial data, there is a tendency to overestimate the capacity of the smallest nodes available in a fab.

These recurring discrepancies can lead to significantly biased results for specific process node sizes. While some of the biases could be mitigated by analysing process nodes in ranges, (e.g. "20-45nm" or "5nm and below"), such an approach would not allow for meaningful insights into the geographic distribution of specific process nodes (e.g. 28nm logic). Because there seems to be no consistent breakdown of capacity by feature size within each fab, it severely restricts the ability to assess substitutability of manufacturing.

Process technologies

For most types of chips, node density (also referred to as feature size) is not the only, and sometimes not even the most relevant, metric. Process nodes can be differentiated by many characteristics beyond the density of the production process.

Foundries often offer special production technologies for their process nodes that are optimised for specific chip types, such as analog, radio frequency (RF), mixed-signal or power semiconductors. For example, on some production lines, GlobalFoundries offers Bipolar-CMOS-DMOS (BCD)3 process technology specifically for the production of power management chips (GlobalFoundries Inc., 2023[18]). Additionally, Table 1 lists the available process technologies for selected process nodes by Tower Semiconductors.

Table 1. Process Technologies at different process nodes from Tower Semiconductors

Process node	Process technology						
	SiGe	RF SOI	RFCMOS	Logic	BCD	CIS	SiPho
45 nm		✓		✓			✓
65 nm	✓	✓	✓	✓	✓	✓	
130 nm	✓	✓	✓	✓			
180 nm	✓	✓	✓	✓	✓	✓	✓
>=250 nm	✓		✓	✓	✓	✓	

Note: Different process technologies are only offered for specific nodes.
Source: Based on Tower Semiconductor (n.d.[19])

Unfortunately, neither SEMI nor TechInsights consistently track the variety of production technologies available per process node. To assess the geographic distribution (and substitutability) of production such information is indispensable. For example, if a chip design company is seeking a second source for their power management chip built on a 40nm process with BCD technology, another foundry's 40nm process optimised for RF applications will be of little use (TSMC, n.d.[20]).

Furthermore, neither of the datasets provide information on capacity distribution by process technology, meaning that wafer starts per month (WSPM) is not broken down by specific process technologies. For instance, even if a fab offers multiple process technologies for mature logic, analog, power, and memory chips, the majority of wafers could still be produced using a single technology, such as BCD.

Currently, publicly available wafer capacity analyses do not address differences in-production technologies and instead focus on comparing node densities.

Semiconductor material

While silicon wafers (Si) continue to be the dominant material for semiconductor production, other wafer materials, such as GaN and silicon-carbide (SiC) vastly outperform silicon in specific applications, such as for power and RF chips (Ma and Gu, 2021[21]).

Fabs often have multiple production lines for different wafer materials, such as GaN, SiC or silicon-germanium (SiGe). However, based on the data in WFW and TechInsights, it is only possible to determine whether a fab can use different types of wafers. The data do not indicate the share of wafer output by each wafer material.[4]

Geographic coverage

In the current geopolitical context, with increasing concerns about excess capacity in legacy chip production, certain companies have little incentive to clearly communicate technological capabilities, production capacities and utilization rates. This is, for example, the case of Chinese Integrated Device Manufacturers (IDMs) and foundries. The availability of data on production capacity is particularly challenging in the case of China. Thus, data about technological capabilities and manufacturing capacity of fabs in China should be interpreted with caution. For example, there have been instances of commercial datasets being updated only after public media reported about new investments in Chinese fabs (King and Debby, 2023[22]). Consequently, it is likely that actual wafer fabrication capacity in China is higher than estimated in this paper.

Developments in semiconductor technology

Developments in semiconductor technology can add complexity to the data and introduce biases. For example, the rise of advanced packaging is likely to result in further diversification in semiconductor production. There are now application specific approaches for wafer fabrication as well as packaging, with foundries offering complete solutions, covering wafer fabrication and packaging. Therefore, assessing substitutability must extend to beyond front-end processes. If a chip design is based on a foundry's process node and an advanced packaging process, it creates a stronger customer-supplier lock-in. This benefits the foundry, which is why most leading foundries are evolving their business models to become "solution providers" (Trendforce, 2024[23]). Consequently, comparing wafer fabrication capabilities and capacities alone is becoming increasingly inadequate and necessitates further work on the measurement of back-end manufacturing.

Uncertainty of upcoming fabs

There is uncertainty about projected production capacity in future fabs, with some more likely to materialise than others. Due to changes in economic, technological, and strategic considerations, not all upcoming fabs will come into operation. Plans to expand capacity might be put on hold due to market downturns, and greenfield investments are sometimes delayed indefinitely. Hence, the reported data used in the analyses presented here may not reflect the latest developments in the dynamic front-end manufacturing fab expansion cycles and should be merely seen as a snapshot in time – September 2025.

The uncertainty surrounding upcoming fabs is evident when comparing fab-level data across different updates from the data providers. Some fabs previously described as having a high likelihood of starting operations are now either on hold or delayed indefinitely. For instance, expansion plans by ams-OSRAM in Kulim, Malaysia, which began construction in 2022, was scaled back in 2024 (Edison Report, 2024[24]). Although the 2023 update reported a high likelihood of operation in the fourth quarter, data recorded since the third quarter of 2024 no longer projects it to become operational in the near future. Similar patterns are observed with data reporting of Intel's fab in Magdeburg, Germany and Samsung fabs in Pyeongtaek, Korea (Hookway, 2025[25]; Trendforce, 2024[23]).

These and other developments may affect the analyses in section 3, which should be interpreted with these considerations in mind.

Annex Endnotes

[1] Technological evolution can be particularly important for technical components of the taxonomy. For example, the use of ranges instead of precise values for “feature size” could help account for some degree of technological evolution. However, the thresholds of such ranges would still require updates. As such, it is proposed that the data are recorded in precise values whenever possible. Data may nevertheless be shared in ranges to preserve confidentiality.

[2] Feature size is typically measured by the half-pitch of the smallest metal lines in the chip. It reflects the smallest distance between two identical features (like transistors or interconnects) in the logic circuitry; for NAND flash memory NAND density is increasingly described by the number of layers in 3D NAND architectures; DRAM feature size is measured by the smallest dimension of the memory cell, typically the half-pitch of the storage capacitor or the transistor gate in the memory array.

[3] BCD (Bipolar-CMOS-DMOS) is a semiconductor process technology that integrates bipolar transistors, CMOS logic, and DMOS power devices on a single chip, and it is commonly used in applications such as power management ICs, motor drivers and automotive electronics.

[4] As an example, WFF only states that Infineon’s new fabs produce SiC- and GaN-based chips but holds no information about the respective shares of wafer capacity.